A Replication of Low (2005): Self-Insurance in a Life-Cycle Model of Labor Supply and Savings

REMARK Replication Package

Abstract

This REMARK revisits Low's life-cycle model of labor supply and savings in an incomplete-markets environment. The central mechanism is that labor-supply flexibility provides an additional self-insurance margin alongside precautionary saving: households can respond to wage risk not only by adjusting consumption and assets, but also by varying hours worked over the life cycle. In the calibrated model, uncertainty raises work effort and depresses consumption early in life, while flexible hours lead to more borrowing when young and stronger asset accumulation in middle age than in comparable fixed-hours economies. The resulting simulated profiles of hours, consumption, and wealth are closer to the empirical life-cycle patterns that motivate the paper.

Keywords Precautionary saving; Life-cycle labor supply

JEL codes D91; H31

This document is a REMARK-style replication note for Low (2005). Its purpose is to pair a concise summary of the paper with executable code that reproduces the main quantitative comparisons. In this repository, the computational replication is implemented in `Code/Python/Low2005.py`.

```
GitHub Repo:        https://github.com/jzhan476/Low2005

Other Materials:
 Replication code:  Code/Python
```

1 Summary

Low (2005) studies how households self-insure over the life cycle when future wages are uncertain and markets are incomplete. The paper's central point is that labor supply is itself an insurance margin: if households can change hours worked, then they can respond to wage risk using both savings and labor effort rather than savings alone.

This REMARK summarizes the paper briefly while leaving the main computational work to the accompanying code. The Python implementation reproduces the paper's core comparisons between certainty and uncertainty and between flexible and fixed labor supply.

The main economic conclusions are straightforward. Under uncertainty, households work more and consume less early in life. With flexible labor supply, they borrow more when young, accumulate more assets in middle age, and generate hours and consumption profiles that are closer to the empirical life-cycle patterns discussed in the paper. The model belongs to the Bewley-style incomplete-markets tradition (Huggett, 1993, Aiyagari, 1994) and the precautionary-saving literature that grew out of it (Kimball, 1990, Carroll, 1997, Carroll and Samwick, 1997); related life-cycle work on labor supply with risk includes Imai and Keane (2004), Pijoan-Mas (2006), Domeij and Flódén (2006), Erosa et al. (2016), and related empirical and modern context can be found in Heathcote et al. (2010), Chetty (2012), Kaplan et al. (2018).

2 Model Setup

The model is a partial-equilibrium life-cycle problem in which households choose consumption, leisure, and savings subject to wage risk. Period utility takes the Cobb–Douglas CRRA form

$$u(c_t, l_t) = \frac{\left(c_t^{\eta} l_t^{1-\eta}\right)^{1-\gamma}}{1-\gamma}. \tag{1}$$

Assets evolve according to

$$A_{t+1} = (1+r)\left[A_t + (H - l_t)w_t - c_t\right], \tag{2}$$

where $H - l_t$ is hours worked. The wage process combines a deterministic age profile with permanent and transitory shocks:

$$\ln w_t = \ln w_0 + \alpha_1 t + \alpha_2 t^2 + \ln w_t^P + \nu_t, \tag{3}$$

$$\ln w_t^P = \ln w_{t-1}^P + \varepsilon_t. \tag{4}$$

The key mechanism is that uncertainty affects both consumption and labor choices. If wages are flexible and labor supply is adjustable, the household can react to bad outcomes by changing hours, which lowers the utility cost of building precautionary balances. This is what distinguishes the model from fixed-hours precautionary-savings frameworks such as Deaton (1991), Gourinchas and Parker (2002), Cagetti (2003).

3 Calibration and Computation

The paper calibrates the model at annual frequency and chooses the discount factor and the utility weight on leisure to match average male hours and median wealth accumulation over working life. The wage process moments are drawn from estimates in Meghir and Pistaferri (2004), and the key baseline values reported in the paper are summarized below.

Table 1 Baseline calibration: common parameters reproduced from Low (2005), Table 1, p. 956, and the headline discount rate $\delta = 0.032$ used for the uncertainty + flexible-hours scenario.

Parameter	Value
Real interest rate r	0.016
Relative risk aversion γ	2.2
Consumption share η	0.4
Wage-growth coefficient α_1	0.0561
Wage-growth coefficient α_2	-0.000599
Permanent-shock variance σ_ε^2	0.031
Transitory-shock variance σ_ν^2	0.030
Social-security replacement rate	0.55
Headline discount rate δ	0.032

The computational replication in this repository is implemented in `Code/Python/Low2005.py`. Following the paper, the code focuses on the working-life behavior that drives the main comparative statics: consumption, hours, and assets under certainty versus uncertainty and under flexible versus fixed labor supply. The simulation uses a fixed random seed and the `LaborIntMargConsumerType` and `IndShockConsumerType` life-cycle solvers from the open-source HARK toolkit (Econ-ARK Team, 2024) for the flexible-hours and fixed-hours economies, respectively. Retirement (ages 65–84) is appended as a closed-form deterministic forward extension for plotting only; see Section 6 below.

3.1 Implied Frisch elasticity

The Cobb–Douglas-in-CRRA preferences in Low (2005) deliver a closed form for the Frisch elasticity of labor supply, which is useful both as an internal consistency check on the calibration and as a way to locate the model in the empirical labor-supply literature. Holding the marginal utility of consumption u_c constant and totally differentiating the intratemporal first-order condition $u_z/u_c = w$ for $u(c, z) = (c^\eta\, z^{1-\eta})^{1-\gamma}/(1-\gamma)$ gives

$$\text{Frisch} \;\equiv\; \left.\frac{\partial \log h}{\partial \log w}\right|_{u_c} \;=\; \frac{1-h}{h} \cdot \frac{1+\eta(\gamma-1)}{\gamma}, \tag{5}$$

where $h = 1 - z$ is the labor share. At the calibration target $h = 0.40$ with $\eta = 0.4$ and $\gamma = 2.2$ this yields Frisch ≈ 1.0.[1] This sits above standard cross-sectional micro estimates—Chetty (2012) reports a preferred Hicksian intensive-margin elasticity of ~ 0.33 and a Frisch upper bound of ~ 0.5 from a meta-analysis of micro studies—but is comparable to the lifecycle-MaCurdy estimates that explicitly correct for borrowing constraints in Domeij and Flódén (2006) (whose adjustment roughly doubles the naive Frisch in calibrated economies similar to Low's) and to the indirect-inference estimates of Imai and Keane (2004), which place the lifecycle Frisch above unity once human-capital accumulation is modeled. The position of Low's calibration in the upper micro range is therefore consistent with the macro-style elasticity that the flexible-hours channel needs to do meaningful precautionary work in a quantitative life-cycle model.

Table 2 Sensitivity of the structural Frisch elasticity (equation (5)) to the consumption share η at Low's published hours target $h = 0.40$ and baseline $\gamma = 2.2$. This compares "apples to apples" holding the *same* hours share; in a full counterfactual each η would shift simulated hours and require re-calibration.

Consumption share η	Frisch (at $h = 0.40$, $\gamma = 2.2$)
0.3	≈ 0.93
0.4	≈ 1.01 (baseline Low)
0.5	≈ 1.09

4 Replication targets

Table 3 compares the two working-life summary statistics that `Code/Python/Low2005.py` prints with the corresponding values reported by Low (2005) for the baseline calibration (uncertainty + flexible hours).

Table 3 Replication of Low (2005) baseline calibration targets (uncertainty + flexible hours). Side-by-side values from the paper and from `Code/Python/Low2005.py`.

Statistic (working-life cross-section)	Low (2005)	This replication
Mean hours, fraction of time worked	0.40	0.36
Peak-assets age (full life cycle)	∼60	56

In Low (2005), Table 1, each row reports a calibrated discount factor chosen so that the model matches the PSID median A/y of 1.84 for that specification; in Low's notation the flexible-hours uncertain-wage calibration has $\delta = 0.032$ versus $\delta = 0.028$ for fixed

[1] Algebraically: $(1 - h)/h = 1.5$ and $[1 + \eta(\gamma - 1)]/\gamma = 1.48/2.2 \approx 0.673$, so Frisch $\approx 1.5 \times 0.673 \approx 1.01$.

hours, pairing a modestly higher implied discount rate (lower β) with labour flexibility precisely because adjustable hours absorb part of wage risk directly (cf. Section 5.3). This replication retains a *single* headline $\delta = 0.032$ across the flexible- and fixed-hours comparisons so that the gap in asset profiles isolates labour flexibility rather than a second pass of discount-rate fitting.

The replication is aligned qualitatively with Low (2005), Table 1. Any difference in the two headline numerical targets arises from deliberate limitations of this reproduction (no mortality risk during retirement, no bequest motive, no unemployment-style transitory state). On the qualitative comparisons the script examines (see Section 6), the replication agrees with Low (2005) that uncertainty raises hours early in life and that flexible-hours households display a more pronounced life-cycle saving pattern than fixed-hours households.

5 Key Results

5.1 Flexible labor supply under certainty

Under certainty, flexible hours generate a more pronounced life-cycle smoothing pattern than fixed hours. Young households borrow more, middle-aged households save more, and hours track the deterministic wage profile more closely. In the baseline utility specification, consumption and leisure are Frisch substitutes, so the hump in hours worked induces a corresponding hump in consumption.

5.2 The effect of wage uncertainty

Introducing uncertainty pushes labor supply upward early in life and lowers consumption at young ages. The extra work effort creates a precautionary buffer of assets that can later be used to smooth consumption and reduce hours when uncertainty has partially resolved. In the paper, this mechanism helps explain why observed hours profiles are flatter than the wage profile alone would suggest.

5.3 Flexible versus fixed labor supply under uncertainty

The key comparison in the paper is between economies with uncertain wages but different degrees of labor-supply flexibility. Low (2005) finds that flexibility both changes the shape of asset accumulation and raises the level of precautionary saving in the calibrated baseline, and reports a higher calibrated discount rate when labor supply is flexible ($\delta = 0.032$ versus $\delta = 0.028$ in his Table 1) precisely because flexible-hours agents would otherwise save *more*.

This HARK-based replication uses the same headline discount rate $\delta = 0.032$ (i.e. $\beta = 1/1.032$) for both the flexible- and fixed-hours economies rather than re-calibrating δ separately in each scenario. The substantive replication target is therefore the *shape* of the asset and consumption profiles — in particular, whether the fixed-hours economy accumulates more precautionary wealth than the flexible-hours economy at the common δ. Figure 5 below confirms this ordering: peak assets are substantially higher under fixed hours, consistent with Low (2005)'s finding that flexible hours act as a partial substitute for asset-based self-insurance.

Quantitatively, the HARK simulation prints a peak of mean assets of 1.45 for the flexible-hours economy and 2.85 for the fixed-hours economy at the common discount rate, a roughly twofold gap. The same flexibility effect is identified *across* households in the heterogeneous-agent literature: Pijoan-Mas (2006) shows that endogenous labor supply lowers the welfare cost of idiosyncratic risk by absorbing a sizeable share of income shocks at the intensive margin, and Heathcote et al. (2010) document a similar flexibility effect in a richer environment with permanent and transitory wage components close to those of Meghir and Pistaferri (2004) used here. Erosa et al. (2016) go further by showing that aggregate elasticities consistent with this micro hours response require the very life-cycle composition of households that Low's calibration targets. The twofold gap between flexible- and fixed-hours peak assets in this replication is therefore a clean quantitative reading of a now-standard mechanism: hours act as a state-contingent transfer that economizes on buffer-stock saving in the sense of Carroll (1997) and Carroll and Samwick (1997). The role of *uncertainty* per se is comparably striking: at the same calibration the certainty version of the flexible-hours economy peaks at only 0.16, so the precautionary motive alone roughly accounts for a ninefold increase in working-life peak wealth.

5.4 Alternative preferences

The paper also checks robustness to alternative preferences. The qualitative message is unchanged: labor flexibility continues to matter for life-cycle asset accumulation, while the exact shape of consumption and hours profiles depends on the degree of risk aversion and on the substitutability between leisure and consumption.

1. Additive separability between consumption and leisure preserves the precautionary labor-supply motive, but changes the shape of consumption profiles.
2. Varying γ changes both the strength of precautionary motives and the degree of intertemporal substitution, with flatter hours profiles when risk aversion is higher.
3. Replacing the Cobb–Douglas specification with a CES aggregator shows that the qualitative role of labor flexibility is robust, even though the magnitude of consumption and asset responses depends on the substitutability between leisure and consumption.

6 Replication Figures

The five figures below are generated by `Code/Python/Low2005.py` (and equivalently by `Code/Python/Low2005.ipynb`) from the calibration of Table 1 above and saved to `Figures/`. Working-life profiles (ages 25–64) are produced by HARK's `LaborIntMargConsumerType` and `IndShockConsumerType`, each solved with a deterministic retirement-stage continuation value spliced in as the terminal solution so that working-life agents internalize the retirement-saving motive. Retirement-age profiles (65–84) themselves are appended as a closed-form deterministic forward extension for plotting only – each simulated agent enters retirement with their terminal assets and a constant pension and consumes optimally subject to the Euler equation, a no-borrowing constraint, and no bequests. Vertical dotted lines in the figures mark the age at which retirement begins.

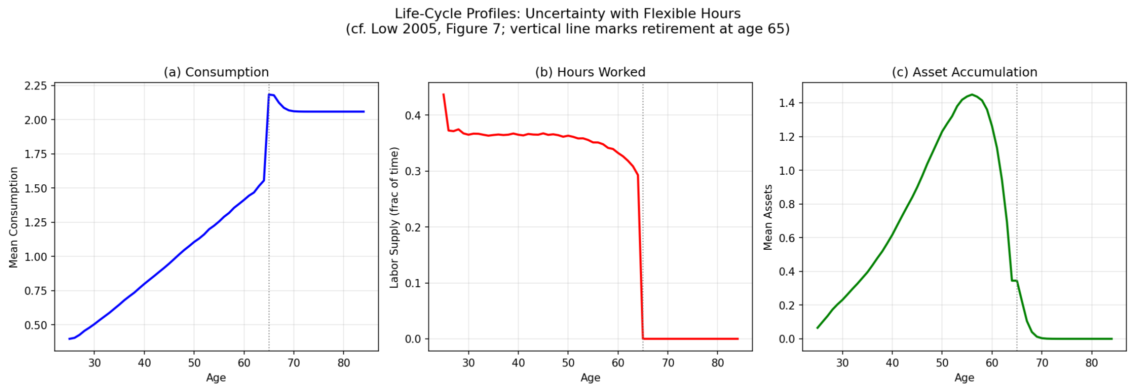

Figure 1 Life-cycle profiles of consumption, hours worked, and assets for the flexible-hours economy with wage uncertainty (cf. Low 2005, Figure 7).

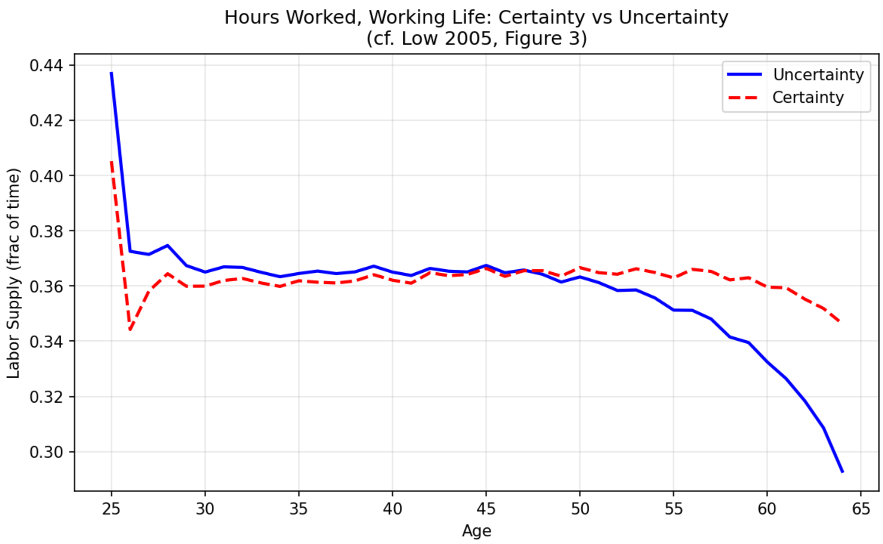

Figure 2 Hours worked over the working life under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 3).

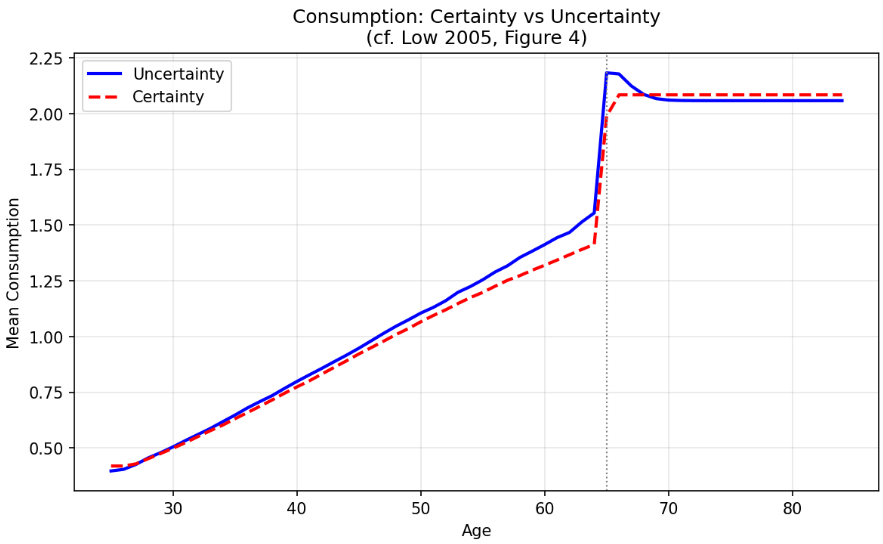

Figure 3 Mean consumption under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 4).

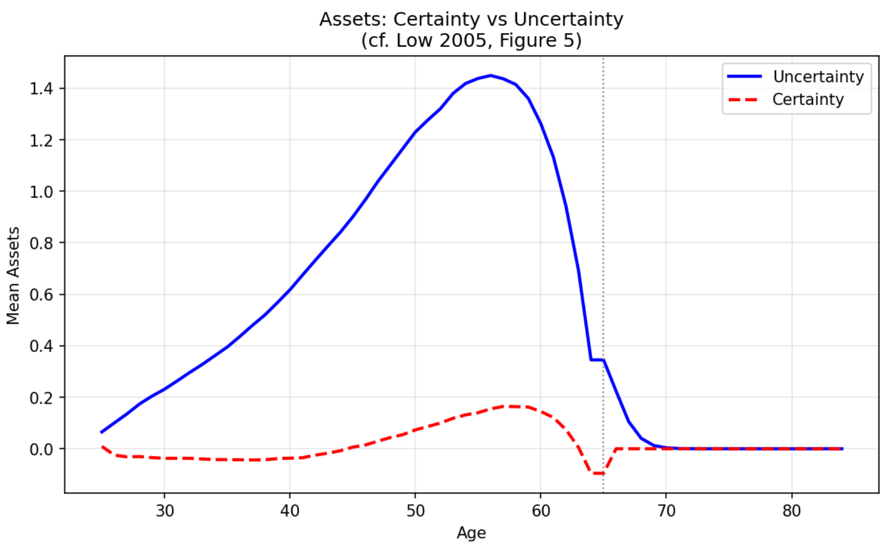

Figure 4 Mean asset holdings under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 5).

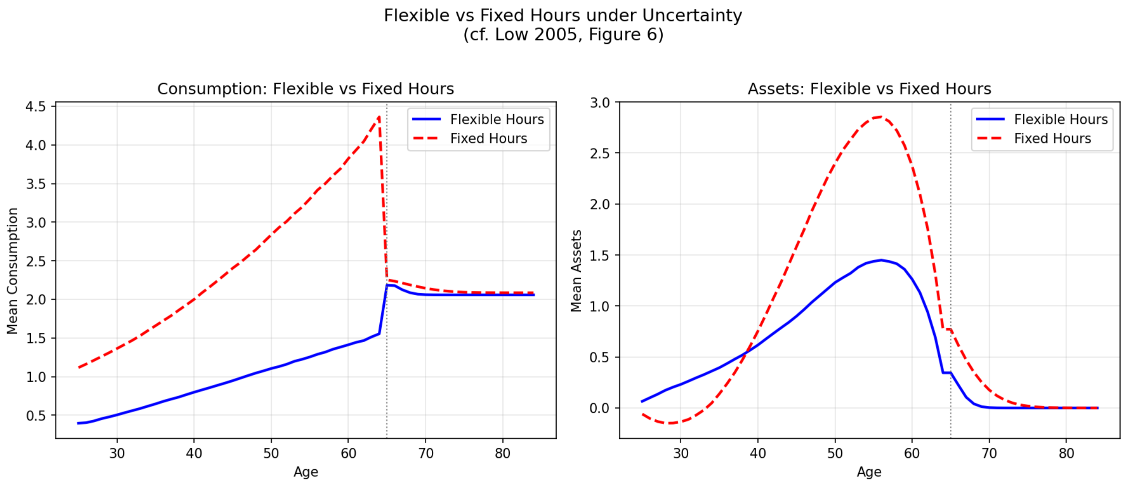

Figure 5 Flexible versus fixed labor supply under wage uncertainty: mean consumption and mean assets over the life cycle (cf. Low 2005, Figure 6).

7 Conclusion

Low (2005) shows that labor-supply flexibility is an important insurance margin in life-cycle models with idiosyncratic wage risk. Ignoring labor supply tends to understate how much of observed wealth is precautionary and misstates the age profiles of work, consumption, and assets. In the calibrated simulations, flexibility leads households to borrow more when young, save more in middle age, and produce life-cycle labor profiles that are closer to the data. This is consistent with later quantitative work such as French (2005), which also benefits from modeling labor supply jointly with savings decisions.

Appendices

A Appendix: Numerical Solution

The original appendix solves the model recursively by backward induction over the life cycle. After using the intratemporal first-order condition to express leisure as a function of consumption and the wage, the dynamic problem can be reduced to solving for the consumption rule on a finite grid of states. In the paper, expectations are evaluated numerically and policy functions are interpolated between grid points. The HARK code in this repository mirrors the economic comparisons of the paper while using modern solution tools for the corresponding life-cycle household problem.

References

S. Rao Aiyagari. Uninsured idiosyncratic risk and aggregate saving. *Quarterly Journal of Economics*, 109(3):659–684, 1994. doi: 10.2307/2118417.

Marco Cagetti. Wealth accumulation over the life cycle and precautionary savings. *Journal of Business & Economic Statistics*, 21(3):339–353, 2003.

Christopher D. Carroll. Buffer-stock saving and the life cycle/permanent income hypothesis. *Quarterly Journal of Economics*, 112(1):1–55, 1997. doi: 10.1162/003355397555109.

Christopher D. Carroll and Andrew A. Samwick. The nature of precautionary wealth. *Journal of Monetary Economics*, 40(1):41–71, 1997. doi: 10.1016/S0304-3932(97)00036-6.

Raj Chetty. Bounds on elasticities with optimization frictions: A synthesis of micro and macro evidence on labor supply. *Econometrica*, 80(3):969–1018, 2012. doi: 10.3982/ECTA9043.

Angus Deaton. Saving and liquidity constraints. *Econometrica*, 59(5):1221–1248, 1991.

David Domeij and Martin Flódén. The labor-supply elasticity and borrowing constraints: Why estimates are biased. *Review of Economic Dynamics*, 9(2):242–262, 2006. doi: 10.1016/j.red.2005.11.001.

Econ-ARK Team. HARK: Heterogeneous Agents Resources and toolKit. Open-source software, https://github.com/econ-ark/HARK, 2024. Econ-ARK project.

Andrés Erosa, Luisa Fuster, and Gueorgui Kambourov. Towards a micro-founded theory of aggregate labour supply. *Review of Economic Studies*, 83(3):1001–1039, 2016. doi: 10.1093/restud/rdw010.

Eric French. The effects of health, wealth, and wages on labour supply and retirement behaviour. *Review of Economic Studies*, 72(2):395–427, 2005.

Pierre-Olivier Gourinchas and Jonathan A. Parker. Consumption over the life cycle. *Econometrica*, 70(1):47–89, 2002.

Jonathan Heathcote, Kjetil Storesletten, and Giovanni L. Violante. The macroeconomic implications of rising wage inequality in the united states. *Journal of Political Economy*, 118(4):681–722, 2010. doi: 10.1086/656632.

Mark Huggett. The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control*, 17(5–6):953–969, 1993. doi: 10.1016/0165-1889(93)90024-M.

Susumu Imai and Michael P. Keane. Intertemporal labor supply and human capital accumulation. *International Economic Review*, 45(2):601–641, 2004. doi: 10.1111/j.1468-2354.2004.00138.x.

Greg Kaplan, Benjamin Moll, and Giovanni L. Violante. Monetary policy according to HANK. *American Economic Review*, 108(3):697–743, 2018. doi: 10.1257/aer.20160042.

Miles S. Kimball. Precautionary saving in the small and in the large. *Econometrica*, 58 (1):53–73, 1990. doi: 10.2307/2938334.

Hamish W. Low. Self-insurance in a life-cycle model of labour supply and savings. *Review of Economic Dynamics*, 8(4):945–975, 2005. doi: 10.1016/j.red.2005.03.002.

Costas Meghir and Luigi Pistaferri. Income variance dynamics and heterogeneity. *Econometrica*, 72(1):1–32, 2004.

Josep Pijoan-Mas. Precautionary savings or working longer hours? *Review of Economic Dynamics*, 9(2):326–352, 2006. doi: 10.1016/j.red.2005.11.003.